

Q1.

Use g as 9.8 m s^{-2} in this question.

Dominic, a bungee jumper of mass 75 kg , has his ankles attached to one end of a cord. The other end of the cord is attached to a bridge which is 50 metres above the surface of a river.

The cord can be modelled as a light elastic cord of natural length 25 metres and modulus of elasticity 3200 N . Dominic is modelled as a particle.

Dominic steps off the bridge at the point where the cord is attached and falls vertically downwards.

- (a) Find Dominic's speed at the point when the cord initially becomes taut. (2)
- (b) Determine whether or not Dominic enters the river and gets wet. (5)
- (c) One limitation of this model is that Dominic is not a particle.
Explain the effect of revising this assumption on your answer to part (b). (2)

(Total 9 marks)

Q2.

In this question use $g = 10 \text{ m s}^{-2}$.

A particle, of mass 2 kg , is attached to one end of a light elastic string of natural length 0.2 metres .

The other end of the string is attached to a fixed point O .

The particle is pulled down and released from rest at a point 0.6 metres directly below O .

The particle then moves vertically and next comes to rest when it is 0.1 metres below O .

Assume that no air resistance acts on the particle.

- (a) Find the maximum speed of the particle. (6)
- (b) Describe one way in which the model you have used could be refined. (1)

(Total 7 marks)

Q3.

In this question use $g = 9.8 \text{ m s}^{-2}$.

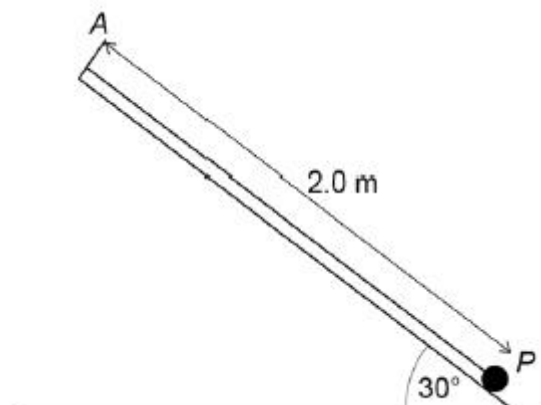
A light elastic string has one end attached to a fixed point, A , on a rough plane inclined at 30° to the horizontal.

The other end of the string is attached to a particle, P , of mass 2 kg .

The elastic string has natural length 1.3 metres and modulus of elasticity 65 N.

The particle is pulled down the plane in the direction of the line of greatest slope through A.

The particle is released from rest when it is 2 metres from A, as shown in the diagram.



The coefficient of friction between the particle and the plane is 0.6

After the particle is released it moves up the plane.

The particle comes to rest at the point B, which is a distance d metres from A.

(a) Show that the value of d is 1.4.

(7)

(b) Determine what happens after P reaches the point B.

Fully justify your answer.

(3)

(Total 10 marks)

Q4.

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(a) An elastic string has natural length l and modulus of elasticity λ . The string is stretched from length l to length $l + e$.

Show, by integration, that the work done in stretching the string is $\frac{\lambda e^2}{2l}$.

(3)

(b) A particle, of mass 5 kg, is attached to one end of a light elastic string. The other end of the string is attached to a fixed point O.

The string has natural length 1.6 m and modulus of elasticity 392 N.

(i) Find the extension of the string when the particle hangs in equilibrium.

(2)

(ii) The particle is pulled down to a point A, which is 2.2 m below the point O.

Calculate the elastic potential energy in the string.

(3)

- (iii) The particle is released when it is at rest at the point A .

Calculate the distance of the particle from the point A when its speed first reaches 0.8 m s^{-1} .

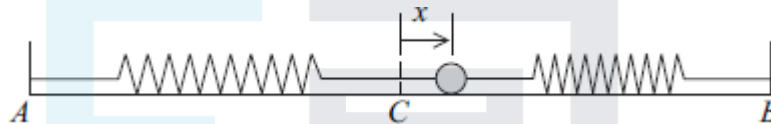
(5)

(Total 13 marks)

Q5.

A particle, of mass 9 kg , is attached to two identical springs. The other ends of the springs are attached to fixed points, A and B , which are 1.2 metres apart on a smooth horizontal surface. The springs have modulus of elasticity 45 N and natural length 0.4 m .

The particle is released from rest at a distance of 0.5 metres from B and moves on the line AB . The midpoint of AB is C . At time t seconds after release, the displacement of the particle from C is x metres, where the direction from A to B is taken to be positive.



- (a) Show that the resultant force on the particle, at time t , is $-225x$ newtons.

(4)

- (b) Hence show that the particle moves with simple harmonic motion.

(2)

- (c) State the period of this motion.

(2)

- (d) Find the speed of the particle when it is 0.05 metres from C .

(4)

- (e) Write down an expression for x in terms of t .

(2)

(Total 14 marks)

Q6.

In this question, take $g = 10 \text{ m s}^{-2}$.

A bungee jumper has mass 75 kg . He uses an elastic rope of natural length 12 m and modulus of elasticity 450 N . He falls vertically, and when the rope becomes taut for the first time, he is travelling at 12.5 m s^{-1} . Assume that, once the rope is taut, the bungee jumper experiences an air resistance force that has magnitude $15v$ newtons, where $v \text{ m s}^{-1}$ is his speed.

At time t seconds after the rope has become taut, the extension of the rope is x metres.

- (a) Show that

$$10 \frac{d^2x}{dt^2} + 2 \frac{dx}{dt} + 5x = 100$$

(4)

- (b) Find x in terms of t .

(10)

- (c) Find the value of t when the bungee jumper first comes instantaneously to rest.

(5)

(Total 19 marks)

Q7.

An elastic string has one end attached to a point O fixed on a rough horizontal surface. The other end of the string is attached to a particle of mass 2 kg. The elastic string has natural length 0.8 metres and modulus of elasticity 32 newtons.

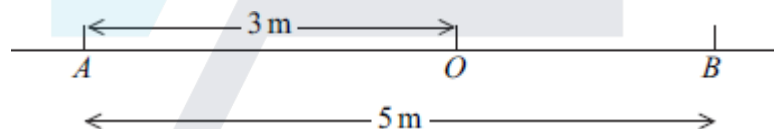
The particle is pulled so that it is at the point A , on the surface, 3 metres from the point O .

- (a) Calculate the elastic potential energy when the particle is at the point A .

(3)

- (b) The particle is released from rest at the point A and moves in a straight line towards O .

The particle is next at rest at the point B . The distance AB is 5 metres.



Find the frictional force acting on the particle as it moves along the surface.

(6)

- (c) Show that the particle does not remain at rest at the point B .

(2)

- (d) The particle next comes to rest at a point C with the string slack.

Find the distance BC .

(2)

- (e) Hence, or otherwise, find the total distance travelled by the particle after it is released from the point A .

(1)

(Total 14 marks)

Q8.

A particle P , of mass 0.4 kg, is attached to one end of a light elastic string, and the other end of the string is attached to a fixed point A .

- (a) The particle P hangs in equilibrium at a point E , vertically below A , where the

extension of the string is 0.2 metres. Calculate the stiffness of the string.

(3)

- (b) The particle P is pulled vertically downwards from the point E by a distance of 0.1 metres, and released from rest. The displacement of P from E at time t seconds after being released is x metres.

- (i) Given that the string does not become slack during the subsequent motion, show that

$$\ddot{x} = hx$$

where h is a constant to be determined.

(4)

- (ii) Hence deduce that the motion of P is simple harmonic.

(1)

- (iii) Show that the period of this motion is 0.898 seconds, correct to three significant figures.

(2)

- (iv) Calculate the maximum speed of P during its motion.

(2)

(Total 12 marks)

Q9.

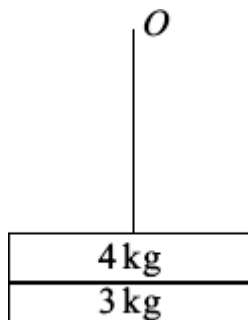
- (a) An elastic string has natural length l and modulus of elasticity λ . The string is stretched from length l to length $l + e$.

Show, by integration, that the work done in stretching the string is $\frac{\lambda e^2}{2l}$

(3)

- (b) A block, of mass 4 kg, is attached to one end of a light elastic string. The string has natural length 2 m and modulus of elasticity 196 N. The other end of the string is attached to a fixed point O .

- (i) A second block, of mass 3 kg, is attached to the 4 kg block and the system hangs in equilibrium, as shown in the diagram.



Find the extension in the string.

(3)

- (ii) The block of mass 3 kg becomes detached from the 4 kg block and falls to the ground. The 4 kg block now begins to move vertically upwards.

Find the extension of the string when the 4 kg block is next at rest.

(6)

- (iii) Find the extension of the string when the speed of the 4 kg block is a maximum.

(3)

(Total 15 marks)

Q10.

A particle, of mass 8 kg, is attached to one end of a length of elastic string. The particle is placed on a smooth horizontal surface. The other end of the elastic string is attached to a point O fixed on the horizontal surface.

The elastic string has natural length 1.2 m and modulus of elasticity 192 N.



The particle is set in motion on the horizontal surface so that it moves in a circle, centre O , with constant speed 3 m s^{-1} .

Find the radius of the circle.

(Total 8 marks)

Q11.

A bungee jumper, of mass 80 kg, is attached to one end of a light elastic cord, of natural length 16 metres and modulus of elasticity 784 N. The other end of the cord is attached to a horizontal platform, which is at a height of 65 metres above the ground.

The bungee jumper steps off the platform at the point where the cord is attached and falls vertically. The bungee jumper can be modelled as a particle. Hooke's law can be assumed to apply throughout the motion and air resistance can be assumed to be negligible.

- (a) Find the length of the cord when the acceleration of the bungee jumper is zero.

(3)

- (b) The cord extends by x metres beyond its natural length before the bungee jumper first comes to rest.

- (i) Show that $x^2 - 32x - 512 = 0$.

(4)

- (ii) Find the distance above the ground at which the bungee jumper first comes to rest.

(4)

(Total 11 marks)

Q12.

- (a) Hooke's law states that the tension in a stretched string of natural length l and modulus of elasticity λ is $\frac{\lambda x}{l}$ when its extension is x .

Using this formula, prove that the work done in stretching a string from an unstretched position to a position in which its extension is e is $\frac{\lambda e^2}{2l}$.

(3)

- (b) A particle, of mass 5 kg, is attached to one end of a light elastic string of natural length 0.6 metres and modulus of elasticity 150 N. The other end of the string is fixed to a point O .

- (i) Find the extension of the elastic string when the particle hangs in equilibrium directly below O .

(2)

- (ii) The particle is pulled down and held at the point P , which is 2 metres vertically below O .

Show that the elastic potential energy of the string when the particle is in this position is 245 J.

(2)

- (iii) The particle is released from rest at the point P . Find the speed of the particle when it reaches O .

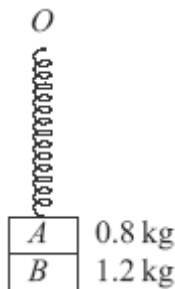
(4)

(Total 11 marks)

Q13.

Two small blocks, A and B , of masses 0.8 kg and 1.2 kg respectively, are stuck together. A spring has natural length 0.5 metres and modulus of elasticity 49 N. One end of the spring is attached to the top of the block A and the other end of the spring is attached to a fixed point O .

- (a) The system hangs in equilibrium with the blocks stuck together, as shown in the diagram.



Find the extension of the spring.

(3)

- (b) Show that the elastic potential energy of the spring when the system is in equilibrium is 1.96 J. (2)
- (c) The system is hanging in this equilibrium position when block B falls off and block A begins to move vertically upwards.

Block A next comes to rest when the spring is **compressed** by x metres.

- (i) Show that x satisfies the equation

$$x^2 + 0.16x - 0.008 = 0$$
 (5)

- (ii) Find the value of x . (2)
- (Total 12 marks)

Q14.

An elastic string of natural length 1.5 metres has one end attached to a fixed point O . A particle of mass 4 kg is attached to the other end of the string. The particle is released from rest at O .

- (a) Find the kinetic energy of the particle when the string becomes taut. (2)
- (b) The particle first comes to rest when it is 3.5 metres below O .

Show that the modulus of elasticity of the string is 103 N, correct to three significant figures. (4)

- (c) Find the speed of the particle when it is 2.7 metres below O . (5)
- (Total 11 marks)

Q15.

A particle, of mass 10 kg, is attached to one end of a light elastic string of natural length 0.4 metres and modulus of elasticity 100 N. The other end of the string is fixed to the point O .

- (a) Find the length of the elastic string when the particle hangs in equilibrium directly below O . (2)
- (b) The particle is pulled down and held at a point P , which is 1 metre vertically below O .

Show that the elastic potential energy of the string when the particle is in this position is 45 J. (2)

- (c) The particle is released from rest at the point P . In the subsequent motion, the particle has speed $v \text{ m s}^{-1}$ when it is x metres **below** O .

- (i) Show that, while the string is taut,

$$v^2 = 39.6x - 25x^2 - 14.6$$

(7)

- (ii) Find the value of x when the particle comes to rest for the first time after being released, given that the string is still taut.

(3)

(Total 14 marks)

Q16.

A light spring of natural length 0.4 metres and modulus of elasticity λ newtons lies on a smooth horizontal table. One end of the spring is attached to the table at a point O and the other end is attached to a particle. The particle is held in equilibrium on the table by a horizontal force of magnitude 12.5 newtons. In this position the length of the spring is 0.5 metres.

- (a) Show that $\lambda = 50$.

(3)

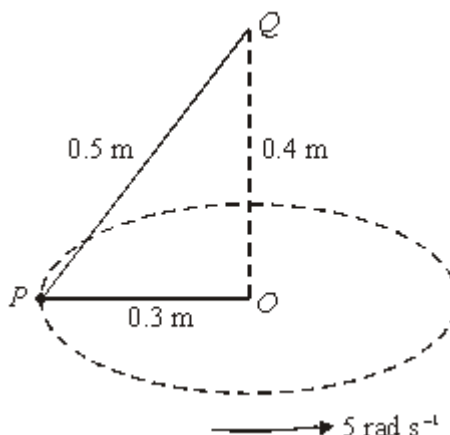
- (b) The horizontal force is removed and the particle, which is of mass 0.2 kg, is released from rest and moves towards O . Calculate its speed when the spring reaches its natural length.

(5)

(Total 8 marks)

Q17.

A particle of mass 0.4 kg is attached at the point P to two light strings, QP and OP . The points O and Q are fixed with Q at a distance of 0.4 m vertically above O . The string QP is inextensible and of length 0.5 m. The string OP is elastic and of natural length 0.2 m and stiffness $k \text{ N m}^{-1}$. The particle moves in a horizontal circle, centre O and radius 0.3 m, at a constant angular speed of 5 rad s^{-1} .



- (a) Draw a diagram showing the forces acting on the particle.

(1)

- (b) Show that the tension in the string QP is 4.9 N. (3)
- (c) Write down, in terms of k , the tension in the string OP . (1)
- (d) Show that $k = 0.6$. (5)
- (e) Find the elastic potential energy stored in the string OP . (2)
- (Total 12 marks)**

Q18.

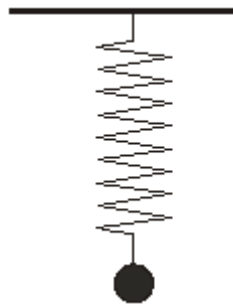
A bungee jumper, of mass 70 kg, is attached to one end of a light elastic cord of natural length 14 metres and stiffness 196 N m^{-1} . The other end of the cord is attached to a bridge, approximately 40 metres above a river.

The bungee jumper steps off the bridge at the point where the cord is attached and falls vertically. The bungee jumper can be modelled as a particle throughout the motion. Hooke's law can be assumed to apply throughout the motion.

- (a) Find the speed of the bungee jumper at the instant the cord first becomes taut. (2)
- (b) The cord extends by x metres beyond its natural length before the bungee jumper first comes instantaneously to rest.
- (i) Show that $x^2 - 7x - 98 = 0$. (4)
- (ii) Hence find the value of x . (2)
- (iii) Calculate the deceleration experienced by the bungee jumper at this point. (4)
- (Total 12 marks)**

Q19.

A sphere of mass 2 kg is fixed to one end of a spring. The other end of the spring is attached to the fixed point P , as shown in the diagram. The natural length of the spring is 0.5 metres.



- (a) When the system is in equilibrium, the length of the spring is 0.7 metres.

Find the modulus of elasticity of the spring.

(3)

- (b) The sphere is pulled down below its equilibrium position and released from rest. At time t seconds, the displacement of the sphere from its equilibrium position is x metres.

- (i) Show that

$$\frac{d^2 x}{dt^2} = -49x$$

(5)

- (ii) Find the value of t when the sphere returns to its equilibrium position for the first time.

(4)

(Total 12 marks)

Q20.

An elastic string has natural length l and modulus of elasticity λ . One end of the string is fixed and a particle of mass m is attached to the other end of the string.

- (a) When the particle hangs vertically in equilibrium, the extension of the string is $\frac{l}{4}$. Show that $\lambda = 4mg$.

(2)

- (b) The particle is then set into vertical motion and x is the displacement of the particle from the equilibrium position at time t . Assume that the string remains taut during the vertical motion.

- (i) Show that $\frac{d^2 x}{dt^2} = \frac{4gx}{l}$.

(5)

- (ii) Hence find the period of the motion.

(2)

(Total 9 marks)

Q21.

An elastic string has natural length 2 metres and modulus of elasticity λ newtons. One end of the string is fixed at the point O , and a particle of mass 20 kg is attached to the other end of the string.

- (a) When in equilibrium the particle is 2.7 metres below O . Show that $\lambda = 560$.

(3)

- (b) The particle is now held at O and released from rest. The maximum length of the string in the subsequent motion is L .

- (i) Show that L satisfies the equation

$$5L^2 - 27L + 20 = 0$$

(5)

- (ii) Find the maximum length of the string.

(3)

(Total 11 marks)

Q22.

A spring, of natural length $4a$ and modulus λ , has one end attached to a fixed support A , and a particle P of mass m is attached to its other end. Another spring, of natural length $2a$ and modulus $4mg$, has one end attached to P and the other end attached to a fixed support B , which is situated at a distance of $10a$ vertically below A . The system is in equilibrium in a vertical line with the upper spring stretched to a length of $7a$ and the lower spring stretched to a length of $3a$ as shown in the diagram.



- (a) Show that $\lambda = 4mg$.

(4)

- (b) At time $t = 0$, the particle is lowered to a distance $\frac{a}{2}$ below its equilibrium position and released from rest. The subsequent motion of P is subject to a resistance of magnitude $\frac{1}{5}mkv$, where $k^2 = \frac{6g}{a}$ and v is the speed of the particle at time t .

- (i) Given that x is the downward displacement of P from its equilibrium position at time t , show that



$$10 \frac{d^2 x}{dt^2} + 2k \frac{dx}{dt} + 5k^2 x = 0$$

(6)

- (ii) Hence find x in terms of a , k and t .

(9)

(Total 19 marks)

Q23.

A particle, of mass m , is attached to one end of a spring. The spring has natural length l and modulus of elasticity λ . The other end of the spring is fixed to the point Q . Initially the particle is in equilibrium, vertically below Q .

- (a) Find, in terms of m , g , λ and l , the extension of the spring when the particle is in equilibrium.

(2)

- (b) The particle is then pulled down, so that it is vertically below its equilibrium position, and released. The displacement of the particle from its equilibrium position is x at time t . Show that

$$\frac{d^2 x}{dt^2} = -\frac{\lambda x}{ml}$$

(5)

- (c) Find, in terms of A , l and m , the period of the subsequent motion.

(2)

- (d) The spring is now replaced with another spring of natural length l , but with modulus of elasticity $\frac{\lambda}{4}$. What happens to the period of the motion?

(2)

(Total 11 marks)

Q24.

A particle P of mass m is suspended from a fixed point O by a light elastic string of natural length a and modulus $6mn^2a$, where n is a positive constant. The particle is released from rest at a point A , where A is vertically below O and distance OA is a .

When the particle is moving with velocity $v \text{ m s}^{-1}$ it experiences air resistance of magnitude $2mnv \text{ N}$.

- (a) The displacement of P below A at time t is x .

Show that x satisfies the equation

$$\frac{d^2 x}{dt^2} + 2n \frac{dx}{dt} + 6n^2 x = g$$

(3)

- (b) Find x in terms of n , g and t .

(10)

- (c) Hence show that the first time the particle comes to rest is when $t = \frac{\pi}{\sqrt{5n}}$.

(4)

(Total 17 marks)



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